

Rank	Mutation	P(High)	Exact mutation in DMS	Other substitutions at position
1	Y131H	0.361	◆ 2L: single → High 2L: 2-mutant → High (+K88R) 2L: 4-mutant → Low (+V90F, T126A, C251Y)	Y131C: 8×, best = High Y131N: 2×, best = Low
2	A24V	0.282	1VH: 2-mutant → Parent (+N189D) 1VH: 2-mutant → Low (+T74I) 2L: 5-mutant → Low (+K64E, F97L, R140C, H155R)	— none —
3	I99V	0.275	1VH: 3-mutant → Low (+N65K, L230M) 1VH: 3-mutant → Low (+E47A, G179C) ◆ 2L: single → High	I99T: 9×, best = High I99N: 5×, best = Low I99F: 1×, best = Low
4	E19K	0.265	◆ 2L: single → High	E19G: 6×, best = Low E19D: 3×, best = Low
5	I23L	0.254	◆ 3VRL: single → High	I23V: 9×, best = Parent I23T: 7×, best = Low I23F: 2×, best = Low I23N: 2×, best = Dead I23S: 1×, best = Parent
6	D49V	0.254	◆ 1VH: single → Low 1VH: 3-mutant → Low (+G13D, R114C) 3VRL: 2-mutant → High (+H210R) 3VRL: 3-mutant → High (+T183A, R256C)	D49G: 9×, best = Low D49E: 2×, best = Low D49Y: 1×, best = Low D49N: 1×, best = Dead D49A: 1×, best = Dead
7	R224S	0.246	2L: 2-mutant → High (+I99T)	R224C: 5×, best = Low R224H: 3×, best = Low R224G: 1×, best = Parent
8	A225T	0.240	◆ 2L: single → Parent	A225V: 2×, best = Parent A225P: 1×, best = Dead
9	L170M	0.238	◆ 3VRL: single → High	L170P: 7×, best = Parent L170Q: 3×, best = Dead
10	T109I	0.236	2L: 2-mutant → Parent (+M229V)	T109A: 5×, best = Parent T109S: 3×, best = Low T109N: 1×, best = Low